A Course by Mónica Vega

GLOSSARY OF LIGHTING TERMS

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Mónica Vega course - Additional Resources

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Light - -
Radiant energy that is capable of producing a visual sensation. The visible portion of the electromagnetic spectrum extends from about 380 to 770 nanometers.

Artificial light - - -
Artificial light is any light source that is not naturally occurring. Artificial Light includes things like flash/strobes, of course, but also any street lights, indoor lighting, or other man-made light sources.

Natural Light - - - -
Natural light is light that is generated naturally, the common source of which is the Sun. This is as opposed to artificial light, which is typically produced by electrical appliances such as lamps.

Bioluminescence -
Light created by the body of a living creature, for example by some sea creatures and insects.

Circadian cicle -
Used to describe the processes in animals and plants that happen naturally during a 24-hour period.

Electromagnetic spectrum (EM) ***
The distribution of energy emitted by a radiant source, arranged in order of wavelength or frequency. Includes gamma-ray, X-ray, ultraviolet, visual, infrared, and radio regions.

Candela **
Unit of light intensity; fundamental dimension of lighting engineering. 1 cd is defined as the light intensity emitted by a monochromatic light source with a radiant power of 1/683 W at 555 nm.

Luminous flux **
Unit: Lumen (lm). The luminous flux expresses the total light power emitted by a light source.

Illuminance **
The illuminance, measured in the units of lux (lx), is the ratio between the luminous flux incident upon an area and the size of that area.

Luminance **
Unit: candela/m2 (cd/m2).
The luminance describes the brightness of a surface that emits light either as a light source or by transmission or reflection. The luminance is defined as the ratio of light intensity to the surface projected perpendicular to the direction of observation.

Reflection **
The ability of surfaces to reflect light. The percentage reflection is known as the reflectance, defined as the ratio of reflected to incident luminous flux. Reflection can be directed or diffuse.

Absorption **
The ability of substances to neither reflect nor transmit light. Its dimension is the absorption ratio, which is defined as the ratio of absorbed luminous flux to incident luminous flux.

Transmission **
The ability of materials to transmit light. The dimension of transmission is transmittance, defined as ratio of transmitted luminous flux to incident luminous flux. Transmission can be directed or diffuse.

Ambient lighting *
Lighting throughout an area that produces general illumination.

Focal glow ***
Focal glow is the follow spot on the modern stage. It is the pool of light at your favorite reading chair. It is the shaft of sunshine that warms the end of the valley. It is candlelight on the face, and a flashlight on a stair... Focal glow draws attention, pulls together diverse parts, sells merchandise, separates the important from the unimportant, helps people see.

Play of brillants ***
Play of brilliants is Times Square at night. It is the eighteenth century ballroom of crystal chandeliers and many candle flames. It is sunlight on a fountain or a rippling brook. It is a cache of diamonds in an opened cave. It is the rose window of Chartres... Play of brilliants excites the optic nerves, and in turn stimulates the body and spirit, quickens the appetite, awakens curiosity, sharpens the wit.

Watts ****
Measurement unit for the electric power consumption of lighting fixtures, or any other appliance that runs with electricity. In lighting, lumens can be compared to miles traveled and watts can be compared to fuel consumption.

Incident angle *
The angle between a ray of light arriving at a surface and a line normal (perpendicular) to that surface.

Correlated color temperature of a light source (CCT) *
The absolute temperature of a blackbody whose chromaticity most nearly resembles that of the light source.

Ingress Protection Rating (IP) **
Ingress Protection rating, a two-digit code that indicates the resistance of a lighting fixture to solid particles and liquids, where higher digits indicate enhanced protection. The first digit indicates protection against solids, and the second indicates the protection degree against liquids.

For example, an IP67 rating indicates a higher degree of protection than an IP54 rating.

Impact Protection (IK) /
The K is from “Kinetic” to differentiate from IP rating is defined in international standards and indicates the level of resistance an electrical enclosure provides against mechanical impact. IK ratings are an important factor when choosing LED lighting which may be fitted in an area that could be subject to rough enviroment, attempted vandalism or other impacts.

LED **
Abbreviation for Light Emitting Diode. An electroluminescent radiator that produces light by recombining chargecarrier pairs in a semiconductor. LEDs produce a narrow-band spectral range. White light is obtained by RGB mixing or luminescence conversion.

OLED *
A thin-film light emitting diode composed of carbon-based material layered between two electrode surfaces and which serves as a diffuse area source.

Incandescence **
Light produced by heating a material; Examples include the glow of a candle, an incandescent filament lamp, a shooting star, etc.

Fluorescence **
Fluorescence is a process in which substances are excited by means of radiation to produce light; the wavelength of the emitted light is always higher than the wavelength of the incident radiation. One of the primary technical applications of fluorescence is in fluorescent lamps where ultraviolet radiation is converted into visible light.

Halogen ****
An improved version of incandescent lamps, where the glowing filament is contained in halogen gas, hence its name. Halogen lamps are around 25% more efficient than their incandescent counterparts.

* ANSI / IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering.

** EERCO Glossary

*** Light Matters: Richard Kelly, The Unsung Master Behind Modern Architecture's Greatest Buildings.

**** FLOS Glossary.

***** IDA Glossary.

-- Lighting Research Center Glossary

- Cambridge Dictionary

- - - Slr Lounge

-- - - Designing buildings wiki

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